Aolan Li

aolan.li@yale.edu - Google Scholar - ORCID - Personal Website

PROFESSIONAL SUMMARY

Statistician and quantitative researcher with a Ph.D. in Statistics, an M.S. in Biostatistics, and prior FDA CDER ORISE experience. Expertise in Bayesian model comparison, marginal likelihood and Monte Carlo computation, simulation, clinical-trial analysis, longitudinal methods, and reproducible biomedical research.

EDUCATION

Ph.D. in Statistics, University of Connecticut — **Storrs, Connecticut, USA**
Advisor: Prof. Ming-Hui Chen — *August 2020 – August 2025*

M.S. in Biostatistics, University of Connecticut — **Storrs, CT** *August 2018 – May 2020*

B.S. in Applied Statistics, Sun Yat-sen University (SYSU) — **Guangzhou, China** *August 2014 – May 2018*

TECHNICAL EXPERTISE

Bayesian & Simulation Methods: Bayesian inference; MCMC; Bayesian model comparison; marginal likelihood estimation; Monte Carlo computation; simulation-based power evaluation; BART

Clinical & Longitudinal Methods: clinical-trial design and analysis; repeated measures; LMM/GLMM/MANOVA/MMRM; longitudinal and multivariate models; covariance-structure assessment; missing data; survival analysis

Programming & Reproducibility: R, Python, SAS, Stan, SQL; R Markdown, Shiny; Git; HPC/SLURM

Additional Methods: causal inference and mediation; SEM; regularization; tree-based machine learning; clustering; microbiome and multi-omics analysis

PROFESSIONAL EXPERIENCE

Program Administrator

Yale University School of Nursing, Orange, CT, U.S. — *September 2025 – Present*

- Lead biostatistical work across NIH-funded studies in neurodevelopment, microbiome science, and symptom research, partnering with investigators and clinicians to define endpoints, analysis plans, and interpretable results.
- Build reproducible R pipelines for longitudinal modeling, microbiome differential-abundance analysis, and biomarker-symptom association studies.
- Deliver publication-ready tables, figures, sensitivity analyses, and statistical methods and results for peer-reviewed and collaborative research outputs.

ORISE Statistical Research Intern

U.S. Food and Drug Administration (FDA), CDER, Silver Spring, MD — *May 2023 – August 2023*

- Developed Bayesian and frequentist simulation workflows for oncology clinical-trial design and analysis questions, including imaging-window impacts, treatment switching, and survival-data analysis.
- Implemented end-to-end analysis scripts with clear documentation and reproducible outputs for internal review and knowledge transfer.

Graduate Assistant (Statistics, Nursing, and COR^{2}E)

University of Connecticut, Storrs, CT, U.S. — *August 2021 – May 2025*

- Led analyses for interdisciplinary biomedical and social-science studies and produced publication-ready tables, figures, sensitivity analyses, and statistical write-ups.
- Modeled longitudinal outcomes and trajectories using LMM/GLMM and functional regression, and conducted simulation-based power studies and high-dimensional variable selection.
- Applied causal mediation, machine learning, and reproducible microbiome and multi-omics workflows to clinical and observational research questions.

SELECTED BAYESIAN AND CLINICAL-TRIAL RESEARCH

- **Longitudinal model assessment:** Dissertation, *Advancing Marginal Likelihoods for Longitudinal Data Analysis*, investigated marginal-likelihood decomposition for covariance-structure assessment in repeated-measures randomized controlled trials.
- **Monte Carlo methodology:** First author of a peer-reviewed comparison of Monte Carlo-based marginal likelihood estimators for Bayesian computation and model comparison.
- **Historical data and missingness:** Presented a Bayesian approach for longitudinal analysis leveraging multiple historical data sources in the presence of missing data at the 2022 Eastern Asia Chapter of the International Society for Bayesian Analysis meeting.

PEER-REVIEWED JOURNAL ARTICLES

1. Chen, J., **Li, A.**, Wu, W., Xu, W., Zhao, T., Starkweather, A. R., Rodriguez, L., Chen, M. H., & Cong, X. S. (2026). Gut microbiota signatures differentiate trajectory-defined response phenotypes and predict self-management outcomes in irritable bowel syndrome. *Frontiers in Microbiomes, 5*, 1884540.
2. **Li, A.**, Liu, P. Y., Wang, Y. B., Milkey, A., Lewis, P. O., & Chen, M. H. (2026). A comparison of Monte Carlo based marginal likelihood estimators. *Wiley Interdisciplinary Reviews: Computational Statistics, 18*(1), e70058.
3. Zhao, T., **Li, A.**, Chen, M. H., Matson, A., & Cong, X. S. (2026). Early-life painful and stressful exposures and neurodevelopment in preterm infants. *Frontiers in Pediatrics, 14*, 1820878.
4. Wu, W., **Li, A.**, Zhao, T., Chen, J., Lainwala, S., Matson, A. P., Chen, M. H., & Cong, X. (2026). The longitudinal impact of early relational contact in the NICU and post-discharge childcare quality on preterm infants' behavioural development up to 18–24 months of corrected age. *Infant and Child Development, 35*(2), e70090.
5. Graziano, T. A., **Li, A.**, Oppong, A. F., Cousin, L., Kelly, D. L., Starkweather, A., & Lyon, D. E. (2026). Racial disparities in psychoneurological symptoms and health-promoting behaviors among breast cancer survivors: A two-year longitudinal study. *Oncology Nursing Forum, 53*(1), 1–15.
6. Zhao, T., **Li, A.**, Wu, W., Chen, J., Combellick, J., Chen, M. H., & Cong, X. (2026). Maternal racial differences and socioeconomic status in preterm infant neurodevelopment, feeding, and growth. *Nursing Research, 75*(2), 88–95.
7. Wu, W., **Li, A.**, Graziano, T. A., Salner, A., Chen, M. H., Singh, V., Judge, M. P., Cong, X., & Xu, W. (2025). Mediating effect of self-efficacy on the relationships among patient-provider partnership, pain, and quality of life in individuals with cancer. *Oncology Nursing Forum, 52*(6), 448–459.
8. Wu, W., Wang, W., **Li, A.**, Chen, J., Lainwala, S., Matson, A. P., Chen, M. H., Li, J., & Cong, X. (2025). Neonatal pain experience and pain sensitivity trajectories in preterm infants: A longitudinal study of flexion withdrawal reflex thresholds over the first two years of age. *The Journal of Pain, 37*, 105558.
9. Wu, W., Chen, J., **Li, A.**, Chen, M. H., Starkweather, A., & Cong, X. (2025). Mechanistic insights into a self-management intervention in young adults with irritable bowel syndrome: A pilot multi-omics study. *Biomedicines, 13*(9), 2102.
10. Wu, W., **Li, A.**, Singh, V., Salner, A., Chen, M. H., Judge, M. P., & Xu, W. (2025). Pain, fatigue, and associated gene expressions over chemotherapy in patients with colorectal cancer. *PLOS ONE, 20*(6), e0325849.
11. Zhao, T., **Li, A.**, Chang, X., Xu, W., Quinn, T., Chen, J., & Cong, X. (2025). Sex differences in mothers' own milk and neurodevelopmental outcomes in preterm infants. *Frontiers in Pediatrics, 13*, 1523952.
12. McNaboe, R. Q., Kong, Y., Henderson, W. A., Cong, X., **Li, A.**, Seo, M. H., & Posada-Quintero, H. F. (2025). Optimizing sensor locations for electrodermal activity monitoring using a wearable belt system. *Journal of Sensor and Actuator Networks, 14*(2), 31.
13. Xu, W., **Li, A.**, Yackel, H. D., Sarta, M. L., Salner, A., & Judge, M. P. (2024). Dietary consumption patterns in breast cancer survivors: Pilot evaluation of diet, supplements and clinical factors. *European Journal of Oncology Nursing, 72*, 102678.
14. Zhao, T., **Li, A.**, Reese, B., Cong, Q., Corwin, E. J., Taylor, S. N., . . . & Cong, X. (2024). Association between mitochondrial DNA copy number and neurodevelopmental outcomes among Black and White preterm infants up to two years of age. *Interdisciplinary Nursing Research, 3*(3), 149–156.
15. Wang, Y. B., Milkey, A., **Li, A.**, Chen, M. H., Kuo, L., & Lewis, P. O. (2023). LoRaD: Marginal likelihood estimation with haste (but no waste). *Systematic Biology, 72*(3), 639–648.

PREPRINTS

1. Chen, J., **Li, A.**, Wu, W., Xu, W., Zhao, T., Starkweather, A. R., ... & Cong, X. S. (2025). Machine learning to phenotype pain and predict response to pain interventions among young adults with irritable bowel syndrome. *medRxiv*.
2. Milkey, A., Chen, M. H., Wang, Y. B., **Li, A.**, & Lewis, P. O. (2025). The sequential multispecies coalescent. *bioRxiv*.

SELECTED PRESENTATIONS AND ABSTRACTS

1. **Li, A.** (2026). Sex differences in pain, stress, and omics. Invited Scientific Spotlight, *From Discovery to Impact: Accelerating Women's Health Research Through Science, Policy, and Investment*, Society for Women's Health Research and Women's Health Research at Yale, Yale University, June 15, 2026.
2. Zhao, T., Dostova, Y. A., **Li, A.**, Crespo, J., Potts-Thompson, S., Matute-Arcos, E., Prescott, L., Tsai, T., & Taylor, J. Y. (2026). Expanding the role of nurse practitioners in reproductive endocrinology and infertility: Outcomes from an NP-led fertility practice and comparison with U.S. national benchmarks. Accepted for presentation at the *Foundation for Reproductive Medicine Conference (FRMC 2026)*, New York, NY, December 4–6, 2026.
3. **Li, A.** (2022). A new Bayesian approach for analyzing longitudinal data by leveraging multiple historical data sources in the presence of missing data. 6th Eastern Asia Chapter of the International Society for Bayesian Analysis.
4. Judge, M., Xu, W., **Li, A.**, Hayley, Y., Michelle, S., & Salner, A. (2025). Informing educational outreach in oncology: Evaluation of dietary consumption patterns, supplement consumption, and clinical factors in breast cancer survivors. *Journal of the Academy of Nutrition and Dietetics, 125*(10), A27.
5. Zhao, T., **Li, A.**, Graziano, T., Chen, M. H., Matson, A., & Cong, X. (2025). Racial disparity in the association between early-life pain/stress exposure and neurodevelopmental outcomes among Black and White preterm infants. ENRS 37th Annual Scientific Sessions, Philadelphia, PA.
6. **Li, A.** & Graziano, T. (2023). Association of social determinants of health and COVID-19-related behaviors with cervical cancer screening in Hispanic women. *Oncology Nursing Forum, 50*(2), A45–A46.

PROFESSIONAL SERVICE

- Volunteer, Statistics in Pharmaceuticals, 2024 and 2022; World Meeting of the International Society for Bayesian Analysis, 2021.
- Committee Member, New England Statistical Society Statathon, 2022; Volunteer, New England Rare Disease Statistics Workshop, 2022.